JAMES RICHARDSON

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PROFESSIONAL SUMMARY

I'm a research engineer with a passion for distributed systems and machine learning infrastructure. Over the past 12 years, I've worked across academia and industry, though my career path has been unconventional. I moved from hands-on development roles into research-focused positions, and I've led some fairly ambitious technical initiatives along the way. I'm based in Oxford and enjoy working on problems that sit at the intersection of theory and practical implementation.

WORK EXPERIENCE

Research Engineer, Staff Level

Quantum Dynamics Research Ltd

quantumdynamics.io

Oxford, United Kingdom

June 2023 - Present

Architected a novel distributed computing framework for quantum simulations that reduced computation time by 47% across the research pipeline

Led a cross-functional team of 34 software engineers, data scientists, and research mathematicians on a flagship project spanning three continents

Designed and implemented custom GPU acceleration protocols for tensor operations, resulting in significant performance improvements for deep learning models

Published 6 peer-reviewed papers in top-tier conferences while maintaining full-time engineering responsibilities and managing strategic technical direction

Junior Software Engineer

TechVenture Solutions

techventure.net

London, United Kingdom

July 2014 - December 2015

Developed backend microservices for client-facing financial platforms using Python and Go

Managed the complete redesign and deployment of a legacy payment processing system that handled transactions for over 200,000 daily users

Single-handedly oversaw infrastructure scaling during a critical growth phase, implementing Kubernetes orchestration for 12 separate application clusters

Mentored a rotating team of 8-12 graduate interns while balancing full coding responsibilities and contributing to architectural decisions

Senior Research Scientist

Cambridge Applied Innovations

cambridgeinnovations.co

Cambridge, United Kingdom

January 2016 - May 2023

Conducted foundational research in natural language processing and developed novel attention mechanisms that improved model accuracy by 23%

Led prototyping efforts for emerging technologies including federated learning and privacy-preserving machine learning systems

Collaborated with 15 academic institutions on joint research initiatives, securing three separate grants totaling 2.4 million pounds

Implemented automated testing frameworks and CI/CD pipelines that reduced deployment errors by 61% and established quality standards across the research division

EDUCATION

PhD in Computer Science

University of Cambridge

Cambridge, United Kingdom

2011 - 2014

MSc in Advanced Computing

University of Oxford

Oxford, United Kingdom

2009 - 2011

BSc in Computer Science

University of Bristol

Bristol, United Kingdom

2006 - 2009

TECHNICAL SKILLS

Programming Languages: Python, Go, Java, C++, Rust, JavaScript

Machine Learning: TensorFlow, PyTorch, JAX, scikit-learn, XGBoost, Large Language Models

Data Processing: Apache Spark, Hadoop, Apache Airflow, Pandas, NumPy, Polars

Cloud Platforms: AWS, Google Cloud Platform, Microsoft Azure

DevOps and Infrastructure: Kubernetes, Docker, Terraform, Jenkins, GitLab CI/CD

Databases: PostgreSQL, MongoDB, Cassandra, Redis, Elasticsearch

Research Areas: Distributed Systems, Neural Architecture Search, Federated Learning, Quantum Computing, Graph Neural Networks

Other: Jupyter, Git, LaTeX, SQL, REST APIs, gRPC, Protocol Buffers